

```

gled = Pin(1, Pin.OUT)
bled = Pin(3, Pin.OUT)

i = 0

while True:
    if i > 2:
        i = 0

    if i == 0:
        rled.value(touch.value())
        gled.off()
        bled.off()
    elif i == 1:
        gled.value(touch.value())
        rled.off()
        bled.off()
    elif i == 2:
        bled.value(touch.value())
        rled.off()
        gled.off()
    i += 1
    sleep(0.5)

--- End of ttp223tchctrl.py ---

```

The TTP 223 sensor has an inbuilt red LED, which glows when the I/O goes high the moment your finger is near it. This way you can differentiate a faulty sensor from a working one. The sensor has two jumpers marked A and B. If you short jumper B using some solder tin, the sensor works as a latch in toggle mode. You bring your finger near it and the output stays high, then you touch it again to make it low till the next touch, and so on.

Before moving to our next useful sensor, let me introduce another electronics module that will help you reap great benefits from your Pico board. 5V relay modules are devices (magnetic switches) that can switch high voltage or high current devices on and off using a 5V signal from a Pico board. They can be used to control devices such as lights, fans, motors, solenoids, etc. The 5V relay has three high voltage terminals (NC, C, and NO), which connect to the device you want to control. The other side has three low voltage pins (ground, Vcc, and signal) which connect to the Pico board. Using a relay in your projects is super easy — just browse the internet for basics and you're done.

Let's now move to interfacing the next sensor, which is a photosensor. We'll use an analog photosensor to sense voltage inversely proportional to the amount of light falling upon it. Wire the photosensor to your board as shown in Figure 6, and execute the code shown below to read the fluctuating voltage value (marked by S on the sensor)

using the analog to digital converter from GPIO 28 Pico Pin 34. The '-' marked on the sensor is connected to Pico pin 3 GND and the middle VCC to pin 36 3V3(OUT). Based on the light intensity calculated we are turning on an external red/green/blue LED indicating a range. Just try to cover the photoresistor with your finger or cover it fully with folded paper and you'll see a different colour LED turning on. Use a relay module instead of an LED, and you have a light-controlled switch to automatically turn on/off your home equipment:

```

from machine import ADC, Pin
from time import sleep

photoPIN = 28
rled = Pin(0, Pin.OUT)
gled = Pin(1, Pin.OUT)
bled = Pin(2, Pin.OUT)

def rledon(rled,gled,bled):
    rled.on()
    gled.off()
    bled.off()

def gledon(rled,gled,bled):
    rled.off()
    gled.on()
    bled.off()

def bledon(rled,gled,bled):
    rled.off()
    gled.off()
    bled.on()

def aledoff(rled,gled,bled):
    rled.off()
    gled.off()
    bled.off()

def readLight(photoGP):
    photoRes = ADC(Pin(28))
    light = photoRes.read_u16()
    light = round(light/65535*100,2)
    return light

while True:
    try:
        lgtper = 100.00 - readLight(photoPIN)
        print("light: " + str(lgtper) + "%")

        if lgtper < 10.00:
            rledon(rled,gled,bled)

```